

Apply filters to SQL queries

Project description

I work as a security professional at a large organization. Part of my job is to investigate security issues to help keep the system secure. I just recently discovered some potential security issues that involve login attempts and employee machines. I have to use SQL with filters to find important information that allows the company to follow and take actions accordingly.

Retrieve after hours failed login attempts

There was a potential security incident that occurred after business hours which normally end at 18:00. The cybersecurity team needs to look for the failed attempts that occurred during that time. The following screenshot shows how I used SQL to filter the info we needed:

```
MariaDB [organization]> select *  
-> from log_in_attempts  
-> where login_time > '18:00' and success = '0';
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0
28	aestrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0

We have two sides of information; the first side is the query and the second one is part of the output.

In the query I selected all (*) information from the `log_in_attempts` table. Then I used `where` clause and `and` operator to get only the times after 18:00 with the unsuccessful attempts.

To confirm:

`login_time > '18:00'` will filter results that occurred after 18:00

`success = '0'` will filter results that were unsuccessful

Retrieve login attempts on specific dates

A suspicious event occurred on 2022-05-09. It is important to check what happened during that day and the day before so I want to review all login attempts.

The following screenshot shows how I used SQL to filter the info we needed:

```
MariaDB [organization]> select *  
-> from log_in_attempts  
-> where login_date = '2022-05-09' or login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0

We have two sides of information; the first side is the query and the second one is part of the output.

In the query I selected all (*) information from the `log_in_attempts` table. Then I used `where` clause and `or` operator to get only login attempts that happened during 8th and 9th of May, 2022.

To confirm:

`login_date = '2022-05-09'` will filter results that occurred on May 9th, 2022

`login_date = '2022-05-08'` will filter results that occurred the day before May 9th, 2022

Retrieve login attempts outside of Mexico

There's been suspicious activity with login attempts, but the team has determined that this activity didn't originate in Mexico. Now, I need to investigate login attempts that occurred outside of Mexico.

The following screenshot shows how I used SQL to filter the info we needed:

```
MariaDB [organization]> select *  
-> from log_in_attempts  
-> where not country like 'mex%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0

We have two sides of information; the first side is the query and the second one is part of the output.

In the query I selected all (*) information from the `log_in_attempts` table. Then I used `where` clause and `not` to filter countries other than Mexico. It is important to notice that options exist for 'Mexico' and its acronym 'Mex'. I used `like` and the `%` sign to make sure that will take in consideration the two options 'Mex' and 'Mexico' as `%` will look for 'else', even if it is for letters or numbers after the 'Mex' reference.

Retrieve employees in Marketing

My team wants to perform security updates on specific employee machines in the Marketing department located in the East offices. I am responsible for getting information on these employee machines and querying them from the **employees** table. The following screenshot shows how I used SQL to filter the info we needed:

```
MariaDB [organization]> select *
-> from employees
-> where office like 'east%' and department = 'marketing';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267
1088	k865l965m233	rgosh	Marketing	East-157
1103	NULL	randerss	Marketing	East-460

We have two sides of information; the first side is the query and the second one is part of the output. This query shows all employees of the marketing department in the East offices. In the query I selected all (*) information from the **employees** table. Then I used **where** clause and **and** operator to get details Marketing and East.

office like 'east%' will filter results that say 'East' and else.

department = 'marketing' will filter departments that say 'Marketing'.

Retrieve employees in Finance or Sales

My team now needs to perform a different security update on machines for employees in the Sales and Finance departments. I have to create a query that identifies all employees in the Sales or Finance departments.

The following screenshot shows how I used SQL to filter the info we needed:

```
MariaDB [organization]> select *
-> from employees
-> where department = 'sales' or department = 'finance';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlsansky	Finance	South-109

We have two sides of information; the first side is the query and the second one is part of the output. This query shows all employees of the sales and finance departments.

In the query I selected all (*) information from the `employees` table. Then I used `where` clause and `or` operator, not `and` since I need employees belonging either to Sales or Finance departments.

`department = 'sales'` will filter departments that say 'Sales'.

`department = 'finance'` will filter departments that say 'Finance'.

Retrieve all employees not in IT

Your team needs to make one more update to employee machines. The employees who are in the Information Technology department already had this update, but employees in all other departments need it. I have to create a query that identifies all employees except for the ones in the Information Technology department.

The following screenshot shows how I used SQL to filter the info we needed:

```
MariaDB [organization]> select *  
-> from employees  
-> where not department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153

We have two sides of information; the first side is the query and the second one is part of the output. This query shows all employees of the Information Technology department.

In the query I selected all (*) information from the `employees` table. Then I used `where` clause with a `not` on it to make sure I will get all employees not related with IT department.

Summary

I successfully applied filters using SQL queries to get the information the company needs to achieve security reviews, analysis and follow ups. All the information was obtained from the `log_in_attempts` and `employees` tables. I used the operators `and`, `or` and `not` to filter required information. I also had to use wildcard `%` and `like` to filter in the scenarios that words could have certain patterns.